measured properties independently. They are like a pair of magical dice, one thrown in Atlantic City and the other in Las Vegas, each of which *randomly* comes up one number or another, yet the two of which somehow manage always to agree. Entangled particles act similarly, except they require no magic. *Entangled particles, even though spatially separate, do not operate autonomously.*

Einstein, Podolsky, and Rosen set out to show that quantum mechanics provides an incomplete description of the universe. Half a century later, theoretical insights and experimental results inspired by their work require us to turn their analysis on its head and conclude that the most basic, intuitively reasonable, classically sensible part of their reasoning is wrong: the universe is not local. The outcome of what you do at one place can be linked with what happens at another place, even if nothing travels between the two locations—even if there isn't enough time for anything to complete the journey between the two locations. Einstein's, Podolsky's, and Rosen's intuitively pleasing suggestion that such long-range correlations arise merely because particles have definite, preexisting, correlated properties is ruled out by the data. That's what makes this all so shocking.[14]

In 1997, Nicolas Gisin and his team at the University of Geneva carried out a version of the Aspect experiment in which the two detectors were placed 11 kilometers apart. The results were unchanged. On the microscopic scale of the photon's wavelengths, 11 kilometers is gargantuan. It might as well be 11 million kilometers—or 11 billion light-years, for that matter. There is every reason to believe that the correlation between the photons would persist no matter how far apart the detectors are placed.

This sounds totally bizarre. But there is now overwhelming evidence for this so-called *quantum entanglement.* If two photons are entangled, the successful measurement of either photon's spin about one axis "forces" the other, distant photon to have the same spin about the same axis; the act of measuring one photon "compels" the other, possibly distant photon to snap out of the haze of probability and take on a definitive spin value—a value that precisely matches the spin of its distant companion. And that boggles the mind.[8]